

Gaëlle Gilardot

Simulation & Optimization Engineer | CFD · Thermal · Multi-objective Optimization

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SUMMARY Aeronautical engineer specialized in CFD and optimization methods, I have industry experience in liquid cooling for high-power chips, automotive batteries, Formula 1 and HVAC. I turn simulations into decisions: Python, CFD and ML optimization methods that deliver measurable performance gains, and ready to bring that impact to the next challenge.

PROFESSIONAL EXPERIENCE

Simulation & Optimization Engineer - CDD | Corintis – Lausanne, Switzerland — May 2026 – Nov 2026

- Continuously built simulation and optimization tools to find the best-performing microfluidic cold plates for high-power chips.
- Created a complete **AI-driven design optimization process** from an existing **Optuna-based CFD workflow**: optimized Latin Hypercube seeding (**23× better design-space coverage**), **Gaussian-process Bayesian optimization** on parallel asynchronous workers, constraints and stopping criteria.
- Benchmarked Bayesian optimizers (GP), TPE & NSGA-II on Pareto hypervolume: **+10% hypervolume**, −1.5 °C on max temperature.
- Translated **customer specifications** into multi-objective constraints: **−1 to −2.9 °C** versus delivered customer designs.
- Benchmarked **2.5D solver robustness** (laminar vs k-ω SST, ECXML stack-ups) against 3D CHT references : OpenFOAM & Fidelity.
- Developed **AI-assisted Python tools** for the engineering team:
  - **Adaptive power scaler** adjusting chip power maps to meet customer requirements with a new coolant.
  - **DoE pipeline** (meshing, parallel solves) grading manufacturing defects by thermal and pressure-drop impact.

Design & Development Engineer | Johnson Electric – Murten, Switzerland — Jan 2025 – Sep 2025

- Designed HVAC components in CAD and planned prototype test campaigns in line with regulations and test standards.
- Monitored experiments and built **Python scripts** to visualize and analyze large test datasets; detailed technical documentation.

Battery Calibration, CFD Simulation & Optimization Intern | AVL – Graz, Austria — Feb 2024 – Sep 2024

- Calibrated a Li-ion automotive battery model against test data by tuning its electrochemical equations during thermal runaway.
- Trained **RNN surrogate models** on simulation data and validated them (RMSE/NRMSE, overfitting control) to drive a **Python calibration tool** based on a 12-parameter DoE.
- Designed the CAD geometry and ran **CFD simulations** to feed the optimization loop; automated the full simulation workflow (setup, batch runs, post-processing) on a **Linux HPC cluster** and integrated the results into **data pipelines**.
- Wrote the internal technical documentation and co-authored a **scientific article** on the study publication ↗.

Heat Exchanger Design & CFD Simulation Intern | AVL – Graz, Austria — Sep 2022 – Feb 2023

- Studied innovative **TPMS** (triply periodic minimal surface) heat exchangers for racing applications; led **CFD analyses** for a new Formula 1 engine cooling system, balancing thermal performance and aerodynamic constraints.

EDUCATION

MEng, Aeronautical & Mechanical Engineering | EPF Engineering School – Paris, France — 2019 – 2024

First Class Honours; ranked 1st of the 4th-year cohort.

High School Diploma, Engineering Sciences, with high honours, European section | Lycée Pierre Corneille — 2016 – 2019

ACADEMIC PROJECTS

- UAV autopilot design (Master's project, 1 year): led a 10-person team to design a drone for a 1-minute autonomous flight; designed and integrated flight control and guidance avionics.
- Business jet sizing (Dassault Falcon Service, 6 months): flight envelope, mission performance, payload-range and fuel-range diagrams for a 3,800 NM range aircraft.
- Satellite power modeling (CNES, 4 months): MATLAB / Simulink model of battery charge and discharge with solar panels.

SKILLS

- **Programming & Tools:** Python (Optuna, scikit-learn, PyTorch, NumPy, pandas, SciPy) Git, MATLAB, Simulink, Java, Linux, HPC
- **Optimization & Machine Learning:** Gaussian-process surrogates, Bayesian optimization, RNN, TPE/NSGA-II, model validation (RMSE/NRMSE, overfitting), multi-objective & constrained optimization, Design of Experiments, LHS/Sobol sampling, AVL CAMEO
- **CFD & Thermal:** Fidelity, OpenFOAM, ANSYS Workbench, Comsol, ParaView; conjugate heat transfer (CHT), k-ω SST turbulence
- **CAD & FEA:** ANSA, Siemens NX & Teamcenter, CATIA V5, SolidWorks

LANGUAGES & INTERESTS

Languages: French (native), English (fluent, C1, TOEIC 960), German (B2, working knowledge)

Interests: competitive badminton, swimming, classical dance

SOFT SKILLS

Learning agility, problem-solving and creativity, clear oral and written communication, adaptability and proactivity